A **AutoAnalyst AI** AI-powered data analysis

← New analysis

# Data Quality and Summary Analysis Report

completed

Dataset: sales_customer_dataset.csv · Asked: 20/07/2026, 10:10:12

## Analysis plan

✓ **1. Provide an overview of the dataset's structure and basic statistics.**

Use pandas to display the first few rows of the dataset with df.head() and describe() to get basic statistics for numeric columns.

▶ Computed results

✓ **2. Identify potential data quality issues related to missing values.**

Check for missing values using df.isnull().sum() and confirm that there are no missing values as indicated in the metadata.

▶ Computed results

✓ **3. Detect any duplicate rows in the dataset.**

Use df.duplicated().sum() to verify that there are no duplicate rows as indicated in the metadata.

▶ Computed results

✓ **4. Identify any unusual or unexpected values in categorical columns.**

Use pandas' value_counts() on categorical columns like 'Gender' to check for unexpected categories, such as 'Other' in 'Gender'.

▶ Computed results

✓ **5. Identify outliers in numeric columns.**

Use box plots for numeric columns such as 'Unit_Price', 'Gross_Sales', and 'Profit' to visually inspect for outliers using matplotlib.

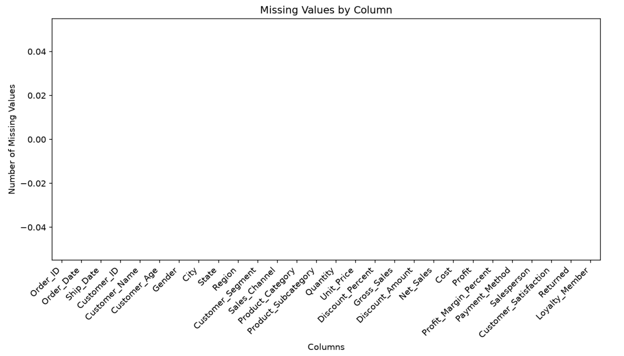
▶ Computed results

✓ **6. Summarize data quality issues found.**

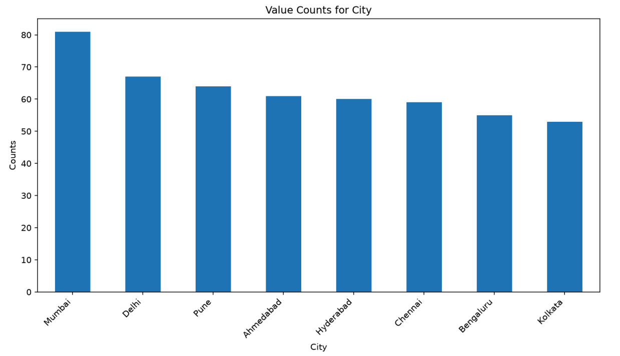
Compile a summary of any data quality issues identified, such as unexpected categories or outliers, and suggest potential actions for data cleaning.

▶ Computed results

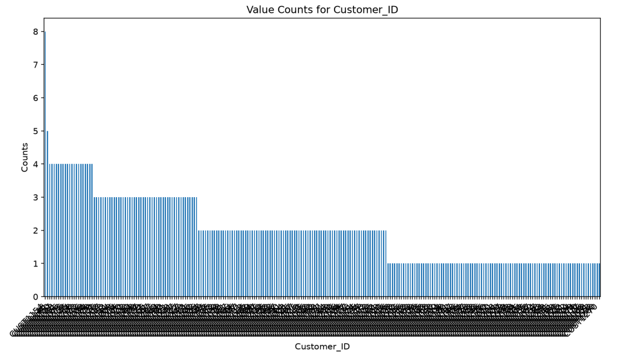
# Charts



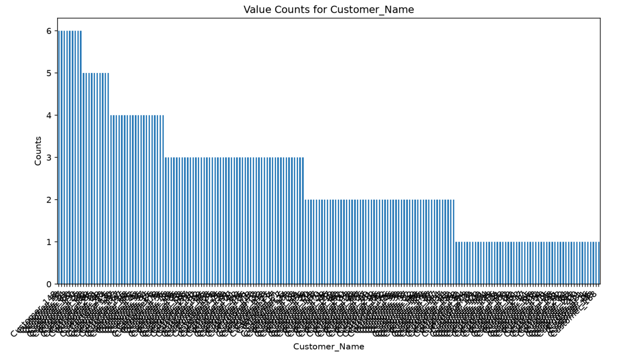

step2_missing_values.png



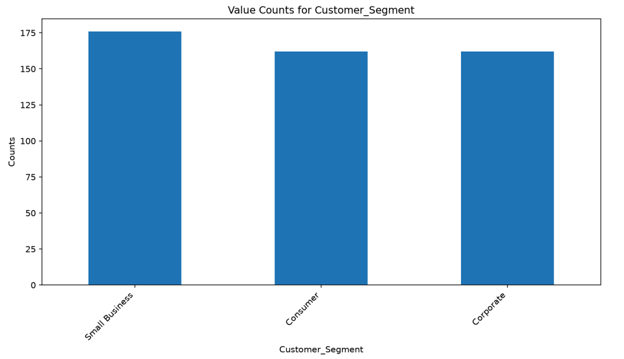

step4_City_value_counts.png



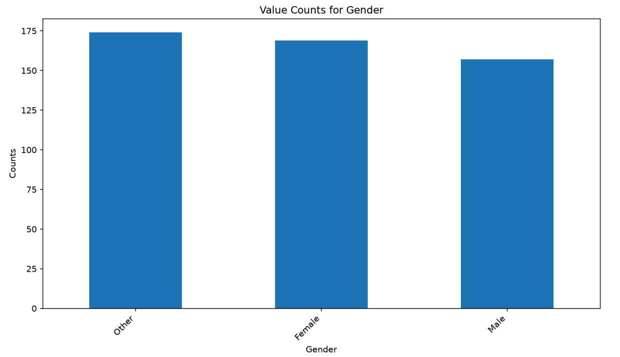

step4_Customer_ID_value_counts.png



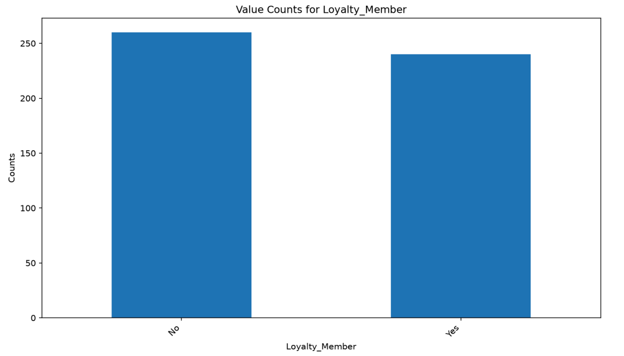

step4_Customer_Name_value_counts.png



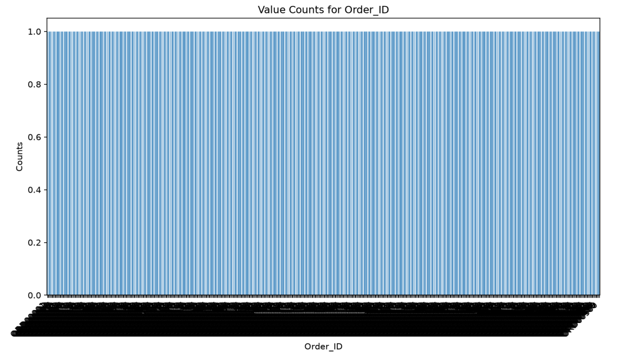

step4_Customer_Segment_value_counts.png



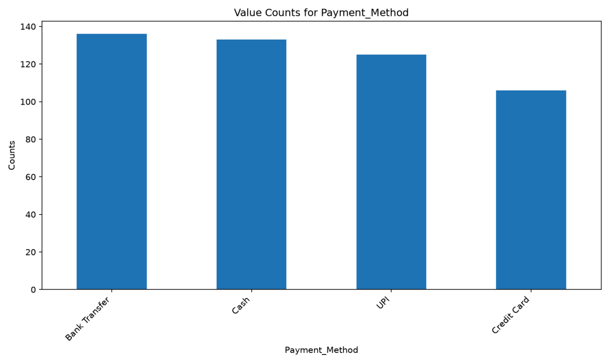

step4_Gender_value_counts.png



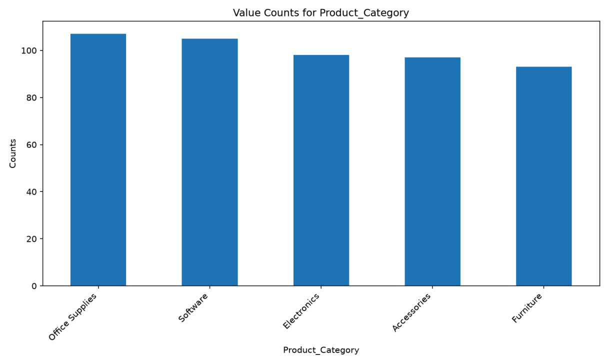

step4_Loyalty_Member_value_counts.png



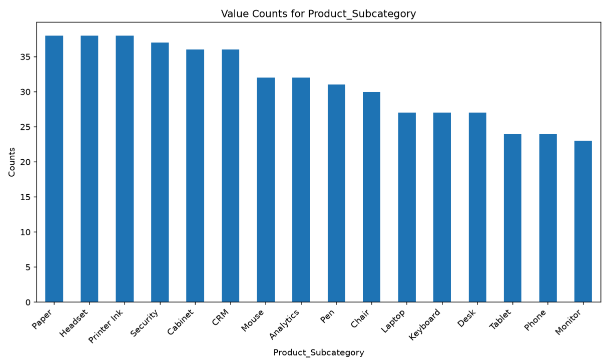

step4_Order_ID_value_counts.png



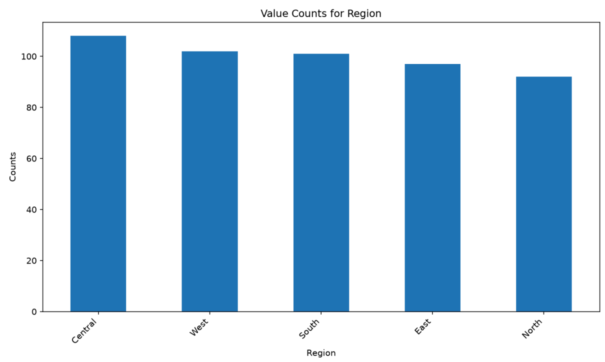

step4_Payment_Method_value_counts.png



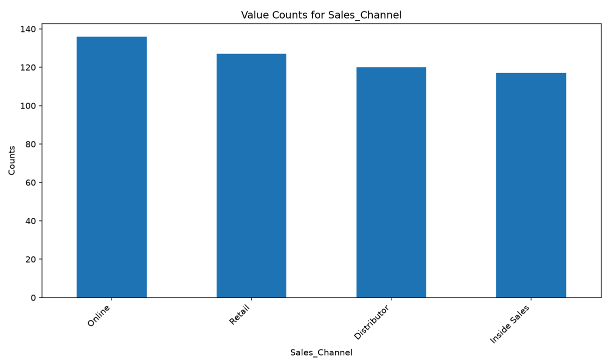

step4_Product_Category_value_counts.png




step4_Product_Subcategory_value_counts.png

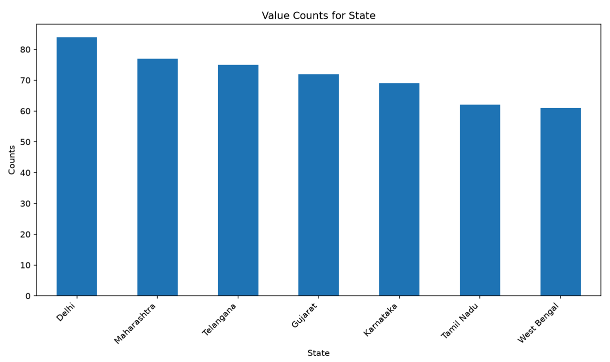



step4_Region_value_counts.png

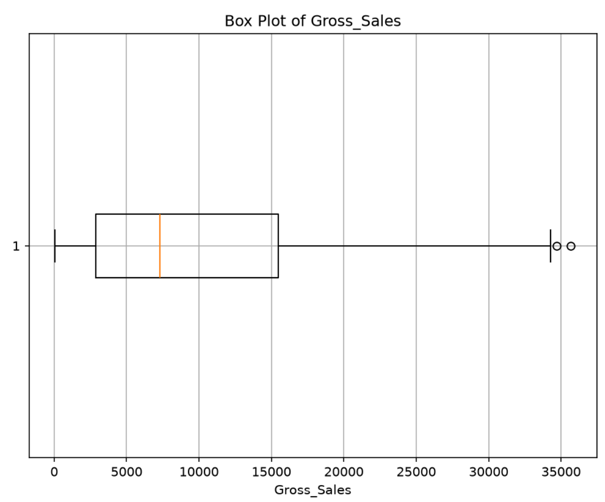



step4_Sales_Channel_value_counts.png

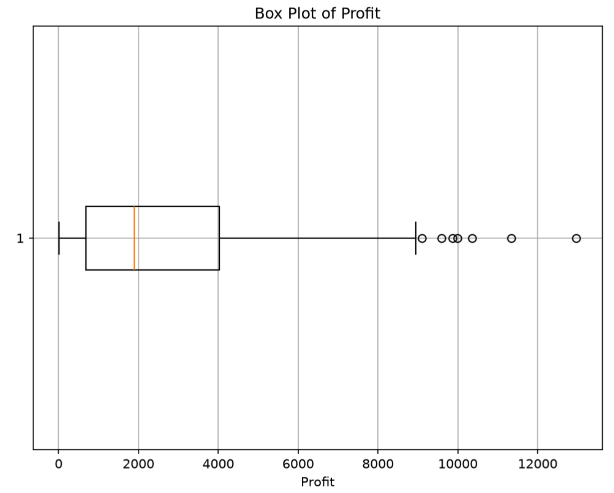

step4_Salesperson_value_counts.png



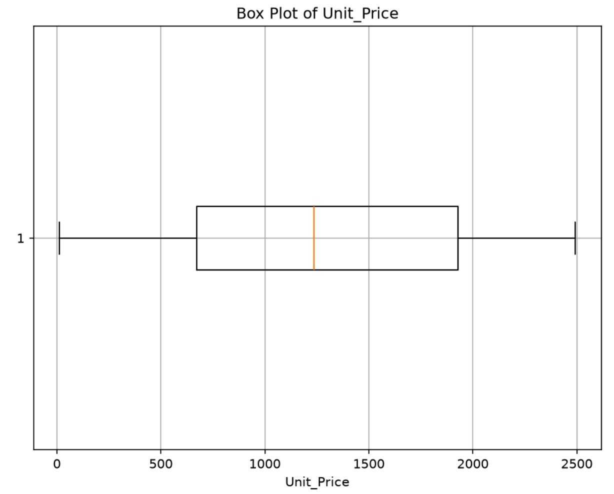

step4_State_value_counts.png



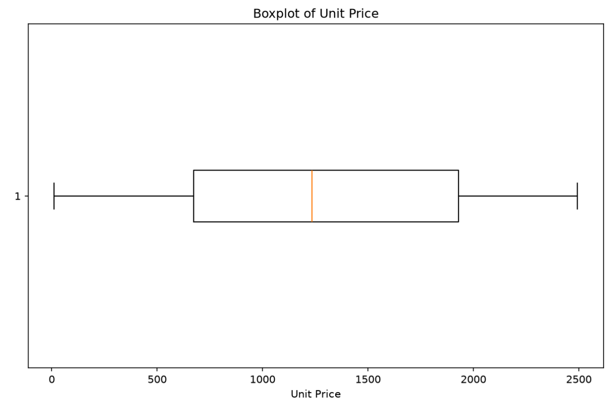

step5_Gross_Sales_boxplot.png




step5_Profit_boxplot.png




step5_Unit_Price_boxplot.png




step6_unit_price_outliers.png

## Insights

### EXECUTIVE SUMMARY

The dataset comprises 500 rows and 28 columns, providing a detailed view of sales transactions. It is complete with no missing or duplicate values, indicating high data integrity. However, there are data quality concerns, including unexpected values in the 'Gender' column and outliers in several numeric columns, which may affect the reliability of the analysis.

### Key findings

The dataset is complete with no missing values across all columns.

📊 Total missing values: 0

There are no duplicate rows in the dataset.

📊 Duplicate rows count: 0

Unexpected value 'Other' found in the 'Gender' column.

📊 Gender unique values: 3, top: 'Other', freq: 174

Outliers detected in numeric columns such as 'Gross_Sales', 'Discount_Amount', and 'Profit'.

📊 Gross_Sales max: 35678.1, Discount_Amount max: 6936.52, Profit max: 12968.3

### Recommendations

- Investigate and address the unexpected 'Other' value in the 'Gender' column to ensure data accuracy.
- Review and potentially adjust outliers in numeric columns to prevent skewed analysis results.

#### Limitations

- The dataset includes future dates (2025), which may not align with current data trends.
- Presence of outliers in several numeric columns could skew analysis results if not addressed.
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